FNCE 30007 DERIVATIVE SECURITIES

Tutorial 3 solution

Forward and Futures pricing

Question 1
A one-year long forward contract on a non-dividend paying stock is entered into when the stock price is $40 and the risk-free rate of interest is 10% per annum with continuous compounding.

a) What are the forward price and the initial value of the forward contract?
b) Six months later, the price of the stock is $45 and the risk-free interest rate is still 10%. What are the forward price and the value of the forward contract?

Answer:

a) The forward price is $F_0 = 40e^{0.1(1)} = 44.21$. The initial value of the contract is zero.
b) The delivery price K is $44.21. The value of the contract f after 6 months is $f = 45 - 44.21e^{-0.1(0.5)} = 2.95$.
The forward price is $45e^{0.1(0.5)} = 47.31$

Question 2
A stock is expected to pay a dividend of AUD 1 per share in two months and in five months. The stock price is AUD 50 and the risk-free rate of interest is 8% per annum with continuous compounding for all maturities. An investor has just taken a short position in a six-month forward contract on the stock.

a) What are the forward price and the initial value of the forward contract?
b) Three months later, the price of the stock is AUD 48 and the risk-free rate of interest is still 8% per annum. What are the forward price and the value of the short position in the forward contract?

Answer:

a) The present value, I, of the income from the security is given by
$I = 1 \times e^{-0.08 \times 2/12} + 1 \times e^{-0.08 \times 5/12} = 1.9540$

From equation (5.2) the forward price, F_0, is given by
$F_0 = (50 - 1.9540)e^{0.08 \times 0.5} = 50.01$
or AUD 50.01. The initial value of the forward contract is (by design) zero. The fact that the forward price is very close to the spot price should come as no surprise. When the compounding frequency is ignored the dividend yield on the stock equals the risk-free rate of interest.

b) In three months
$I = e^{-0.08 \times 2/12} = 0.9868$
The delivery price, K, is 50.01. From equation (5.6) the value of the short forward contract, f, is given by
$f = -(48 - 0.9868 - 50.01e^{-0.08 \times 3/12}) = 2.01$
and the forward price is
$(48 - 0.9868)e^{0.08 \times 3/12} = 47.96$ or AUD 47.96.

Question 3
Suppose that the risk-free rate is 10% per annum with continuous compounding and that the dividend yield on a stock index is 4% per annum. The index is standing at 400, and the futures price for a contract deliverable in four months is 405. What arbitrage opportunities does this create?

Answer:
The theoretical futures price is $400e^{(0.10-0.04)(4/12)} = 408.08$

The actual futures is 405 - the index futures price is too low. Therefore arbitrageur should 1) buy futures 2) short stocks underlying the asset (or short an ETF).

Question 4
The two-month interest rates in Switzerland and the United States are 2% and 5% per annum, respectively, with continuous compounding. The spot price of the Swiss franc is $0.80. The futures price for a contract deliverable in two months is $0.81. What arbitrage opportunities does this create?

Answer:
The theoretical futures price is $0.80e^{(0.05-0.02)(2/12)} = 0.8040$. The actual futures price is too high. The arbitrageur should buy Swiss francs and short Swiss franc futures.

Question 5
Suppose that F_1 and F_2 are two futures contracts on the same commodity with times to maturity, t_1 and t_2, where $t_2 > t_1$. Prove that

$$F_2 \leq F_1 e^{r(t_2-t_1)}$$

where r is the interest rate (assumed constant) and there are no storage costs. For the purposes of this problem, assume that a futures contract is the same as a forward contract.

Answer:
If $F_2 > F_1 e^{r(t_2-t_1)}$ an arbitrageur could
1) long futures that mature at time t1 and
2) short futures that mature at time t2.

When the first contract matures the asset is purchased for F_1 using funds borrowed at rate r. It is then held until time t2 where it is exchanged for F_2 under the second contract. The costs of the funds borrowed at time t2 is $F_1 e^{r(t_2-t_1)}$. A positive profit of $F_2 - F_1 e^{r(t_2-t_1)}$ is realised at time t2. This cannot exist for long and so $F_2 \leq F_1 e^{r(t_2-t_1)}$.

Question 6
When a known future cash outflow in a foreign currency is hedged by a company using a forward contract, there is no foreign exchange risk. When it is hedged using futures contracts, the marking-to-market process does leave the company exposed to some risk. Explain the

nature of this risk. In particular, consider whether the company is better off using a futures contract or a forward contract when:

a) The value of the foreign currency falls rapidly during the life of the contract.
b) The value of the foreign currency rises rapidly during the life of the contract.
c) The value of the foreign currency first rises and then falls back to its initial value.
d) The value of the foreign currency first falls and then rises back to its initial value.

Assume that the forward price equals the futures price.

Answer:
The gain or loss under both contracts is the same however the timing of the cashflows will be different. Once the time value of money is taken into account the futures contract may be more or less valuable than the forward. The long forward provides a perfect hedge while the futures is slightly imperfect.

a) The forward contract is slightly better. The company will make a loss on the hedge. With a forward the whole loss will be realised at the end. Futures the loss is realised each day. On a PV basis the forward is preferable.
b) Futures slightly better. Forward contract the gain is realised at the end. Futures gain is realised day by day. On a PV basis the futures is better.
c) Futures slightly better because positive cashflows early and losses later.
d) Forward slightly better because futures will incur early losses and gains later.

Question 7

The following table (also in excel file T3_Q7_data.xls) contains data on monthly changes in the spot price and the futures price for a certain commodity. Using the approach illustrated in the S&P 500 hedge ratio example, calculate the minimum variance hedge ratio from the given data.

Spot_Change	Futures_Change
0.5	0.56
0.61	0.63
-0.22	-0.12
-0.35	-0.44
0.79	0.6
0.04	-0.06
0.15	0.01
0.7	0.8
-0.51	-0.56
-0.41	-0.46

Answer:

To compute the MVHR, h^*, we need the following quantities:

- *The standard deviation of the spot price change:*
- *The standard deviation of the futures price changes:*
- *The correlation coefficient of the price changes*

We can compute these in Excel to find:

$$\sigma_S = 0.4933$$
$$\sigma_F = 0.5116$$
$$\rho = 0.981$$

Therefore:

$$h^* = 0.981 * \frac{0.4933}{0.5116}$$
$$h^* = 0.946$$